INCH-POUND

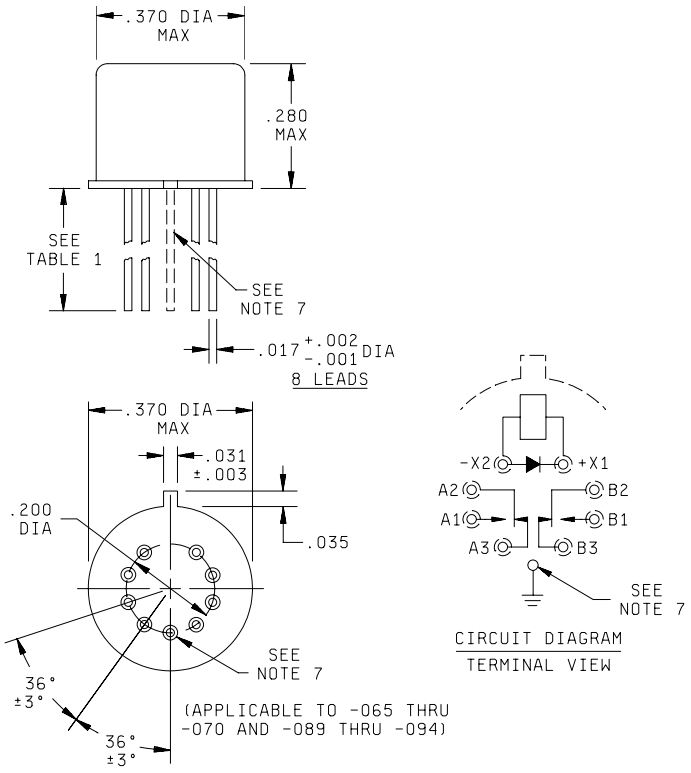
MIL-PRF-39016/15K
3 December 2004
SUPERSEDING
MIL-PRF-39016/15J
13 May 1987

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification sheet and the latest issue of MIL-PRF-39016.

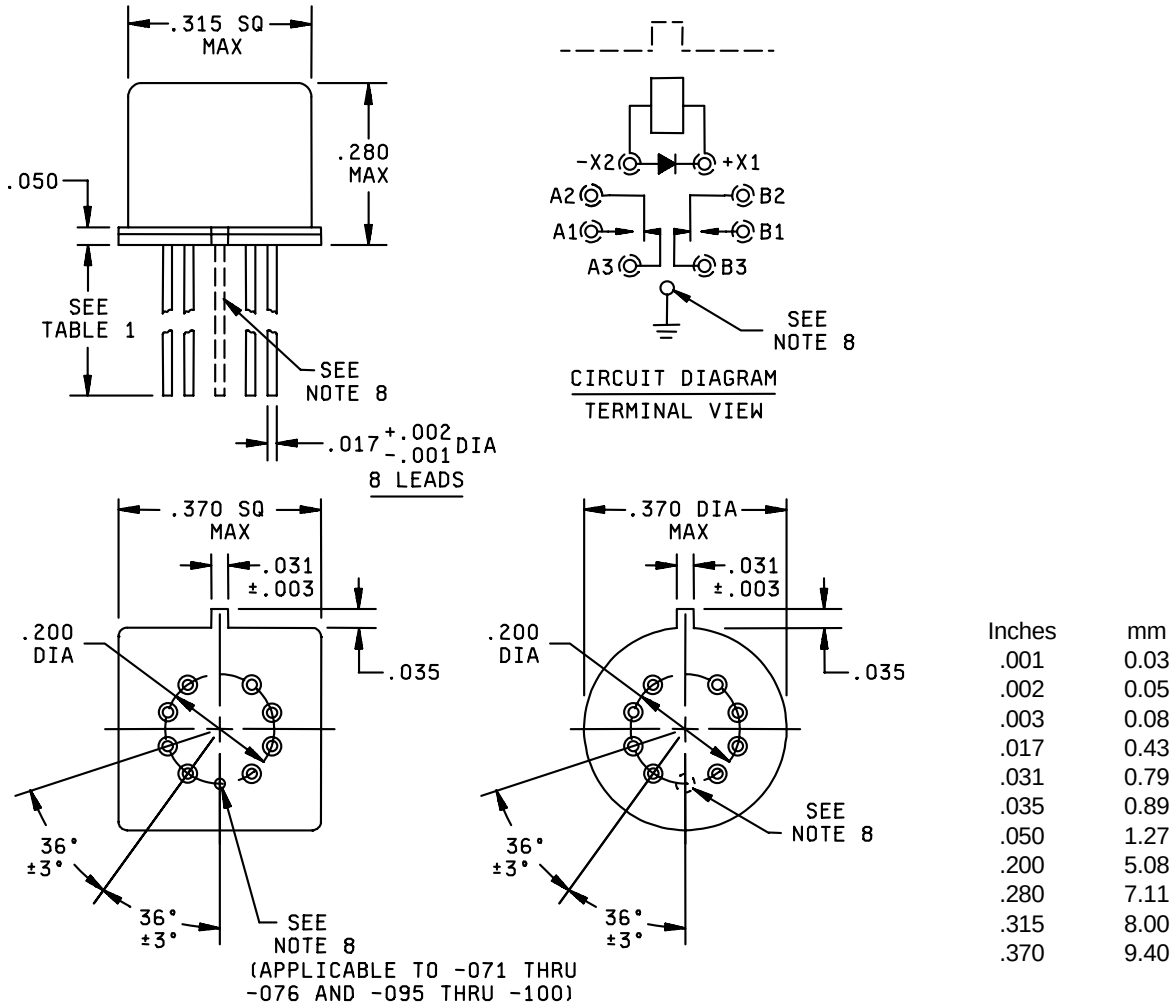


Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.99
.280	7.11
.335	8.51
.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. The grounding pin shown is a noninsulated case ground applicable to -065 through -070 and -089 through -094 (see table I).

FIGURE 1. Dimensions and configuration.

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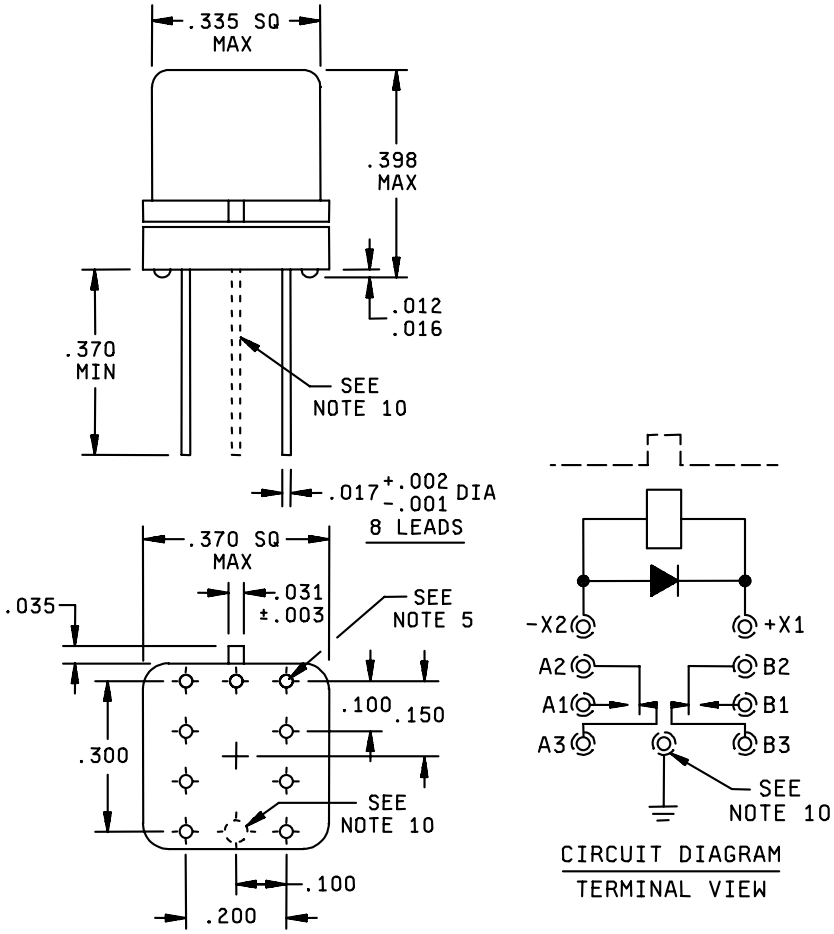
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.050	1.27
.200	5.08
.280	7.11
.315	8.00
.370	9.40

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. Shape optional within the envelope dimensions.
8. The grounding pin shown is a noninsulated case ground applicable to -071 through -076 and -095 through -100 (see table I).

FIGURE 2. Dimensions and configuration (square or round).

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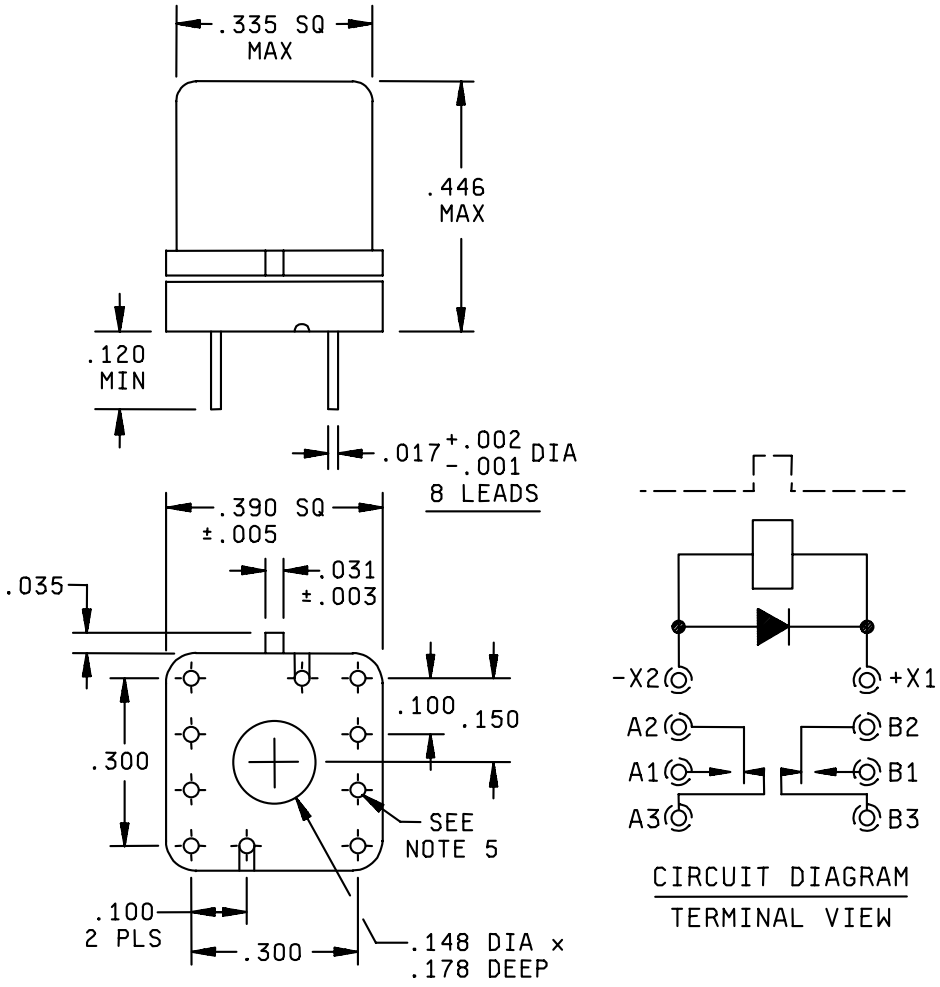


Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader mounting pads shall comply with the requirements of A-A-55485, A-55485/05-003 or A-55485/05-013.
5. Dimensions and tolerances shown for the bottom view of the spreader mounting pad are for the center-to-center locations of the holes in the spreader mounting pad.
6. Shape optional within the envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
9. Circuit diagram shown on part is the terminal view.
10. The grounding pin shown is applicable to dash numbers -113 through -118 only and is a noninsulated case ground (see table I).

FIGURE 3. Dimensions and configuration relay with spreader mounting pad (.100 x .200 terminal spacing) attached (square or round).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader mounting pads shall comply with the requirements of A-A-55485, A-55485/05-014.
5. Dimensions and tolerances shown for the bottom view of the spreader mounting pad are for the center-to-center locations of the holes in the spreader mounting pad.
6. Shape optional within the envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
9. Circuit diagram shown on part is the terminal view.

FIGURE 4. Dimensions and configuration relay with spreader mounting pad (.100 x .300 terminal spacing) attached (square or round).

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:
1.0 ampere at 28 V dc.
250 milliamperes at 115 V ac 60 and 400 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.10 ohm maximum (0.125 ohm maximum with figure 3 spreader mounting pad attached and 0.150 ohm maximum with figure 4 spreader mounting pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.
After life: 0.20 ohm maximum (0.225 ohm maximum with figure 3 spreader mounting pad attached and 0.250 ohm maximum with figure 4 spreader mounting pad attached).

Low level:

During life: 33 ohms maximum.
After life: 0.15 ohm maximum (0.175 ohm maximum with figure 3 spreader mounting pad attached and 0.200 ohm maximum with figure 4 spreader mounting pad attached).

Intermediate current:

During: 1 ohm maximum.
After: 0.20 ohm maximum (0.225 ohm maximum with figure 3 spreader mounting pad attached and 0.250 ohm maximum with figure 4 spreader mounting pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level “L”).

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels “M”, “P”, and “R”).

Overload (high level only): Two times rated current. Not applicable to ac load ratings.

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COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

ELECTRICAL DATA:

Insulation resistance: 1/ 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Post intermediate current life test Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts in the energized and de-energized positions.	500	500	125
Between case, frame, or enclosure and coils.	500	500	All terminals to case
Between all contacts and coils.	500	500	
Between open contacts in the energized and de-energized positions.	500	375	
Between contact poles.	500	500	
Between coils of dual coil relays.	N/A	N/A	

DIODE CHARACTERISTICS: 2/

Maximum negative transient: 1.0 volt.

Coil transient suppression: Applicable.

Semiconductor in process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only.

1/ Connect coil leads together to avoid damage to the diode.
2/ Warning: Reverse polarity on coil terminals will destroy diode.

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/								Coil voltage (V dc) 5/		At 25°C				Over temperature range		
1.500 min leads 3/	1.500 min leads with ground 3/	.187 leads +.040 -.010	.500 min leads with ground	.500 min leads with ground	Spreader mounting pads 4/	Spreader mounting pads with ground 4/	Figure	Rated	Max	Coil resist-ance ohms ±10%	Speci-fied pick-up value (voltage) (V dc)	Speci-fied hold value (voltage) (V dc)	Speci-fied drop-out value (voltage) (V dc)	Speci-fied pick-up value (voltage) (V dc)	Speci-fied hold value (voltage) (V dc)	Speci-fied drop-out value (voltage) (V dc)
029	065	035	077	089	---	---	1	6.0	8.0	98	3.5	2.0	0.28	4.5	3.2	0.18
030	066	036	078	090	---	---		9.0	12	220	5.3	3.0	0.54	6.8	4.9	0.35
031	067	037	079	091	---	---		12	16	390	7.0	4.0	0.63	9.0	6.5	0.41
032	068	038	080	092	---	---		18	24	880	10.5	6.0	0.91	13.5	10.0	0.59
033	069	039	081	093	---	---		26.5	32	1,560	14.2	8.0	1.37	18.0	13.0	0.89
034	070	040	082	094	---	---		5.0	5.8	50	2.7	1.4	0.22	3.5	2.3	0.14
041	071	047	083	095	---	---	2	6.0	8.0	98	3.5	2.0	0.28	4.5	3.2	0.18
042	072	048	084	096	---	---		9.0	12	220	5.3	3.0	0.54	6.8	4.9	0.35
043	073	049	085	097	---	---		12	16	390	7.0	4.0	0.63	9.0	6.5	0.41
044	074	050	086	098	---	---		18	24	880	10.5	6.0	0.91	13.5	10.0	0.59
045	075	051	087	099	---	---		26.5	32	1,560	14.2	8.0	1.37	18.0	13.0	0.89
046	076	052	088	100	---	---		5.0	5.8	50	2.7	1.4	0.22	3.5	2.3	0.14
---	---	---	---	---	101	113	3	6.0	8.0	98	3.5	2.0	0.28	4.5	3.2	0.18
---	---	---	---	---	102	114		9.0	12	220	5.3	3.0	0.54	6.8	4.9	0.35
---	---	---	---	---	103	115		12	16	390	7.0	4.0	0.63	9.0	6.5	0.41
---	---	---	---	---	104	116		18	24	880	10.5	6.0	0.91	13.5	10.0	0.59
---	---	---	---	---	105	117		26.5	32	1,560	14.2	8.0	1.37	18.0	13.0	0.89
---	---	---	---	---	106	118		5.0	5.8	50	2.7	1.4	0.22	3.5	2.3	0.14
---	---	---	---	---	125	---	4	6.0	8.0	98	3.5	2.0	0.28	4.5	3.2	0.18
---	---	---	---	---	126	---		9.0	12	220	5.3	3.0	0.54	6.8	4.9	0.35
---	---	---	---	---	127	---		12	16	390	7.0	4.0	0.63	9.0	6.5	0.41
---	---	---	---	---	128	---		18	24	880	10.5	6.0	0.91	13.5	10.0	0.59
---	---	---	---	---	129	---		26.5	32	1,560	14.2	8.0	1.37	18.0	13.0	0.89
---	---	---	---	---	130	---		5.0	5.8	50	2.7	1.4	0.22	3.5	2.3	0.14

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1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level application.
2/ The suffix letter L, M, P, or R, to designate the applicable failure rate level, shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 029L - - - - - - 118R.
3/ 1.500 leads are inactive for new design.
4/ Relays supplied with spreader mounting pads (-101 through -106, -113 through -118, and -125 through -130) shall have the spreader mounting pad rigidly attached.
5/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

PHYSICAL DATA:

Terminal strength: MIL-STD-202, method 211.

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, ½ pound load.

Twist test: As specified in MIL-PRF-39016.

Solderability: Applicable.

Dimensions and configuration: See figures 1, 2, and 3.

Weight: 2.55 grams (0.09 ounce) maximum, 2.80 grams (0.099 ounce) maximum with spreader mounting pad attached.

Seal: Hermetic.

Minimum marking: Miliatry part number, J date code (example J0430), circuit diagram, manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/15- (dash number from table I and suffix letter designating failure rate level).

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QUALITIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/15-081 or M39016/15-087	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/15-077 M39016/15-078 M39016/15-079 M39016/15-080 M39016/15-082 or M39016/15-083 M39016/15-084 M39016/15-085 M39016/15-086 M39016/15-088	2 units each part number, qualification inspection, Q1.

1/ Figure 1 only - For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-39016/16, /20, and /21 may be used in addition to MIL-PRF-39016/15 data. Qualification to figure 1 automatically qualifies round relays to figure 2.
Figure 2 only - For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-39016/16, /20, and /21 may be used in addition to MIL-PRF-39016/15 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in Q5, MIL-PRF-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection testing the relay manufacturer shall preselect the sample size.

Initial qualification of relays supplied with spreader mounting pads (-101 through -106 and -113 through -118, and -125 through -130) shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Before installation of spreader mounting pad: Screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

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After installation of spreader mounting pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression, solderability, seal, visual and mechanical inspection (external).

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader mounting pad pads (-101 through -106 and -113 through -118, and -125 through -130), two units from figure 3 and two units from figure 4 of the 26.5-volt rated coil voltage (-117 and -129) shall be tested as specified below:

Before installation of pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

For failure rate level L only: Screening.

For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes vibration (random), and screening.

After installation of pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance; specified pickup, hold, and dropout values, coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Figures 3 and 4 only: If the relays produced for MIL-PRF-39016/15 are similar in construction and design except for the diodes and coils, as applicable, to the relays produced for MIL-PRF-39016/20 or MIL-PRF-39016/21, then reduced testing for qualification of MIL-PRF-39016/15 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/20 or MIL-PRF-39016/21.

Group A testing for relays supplied with mounting pad pads (-101 through -106 and -113 through -118, and -125 through -130) shall be tested as specified below:

Perform seal test immediately, preceding the A2 electrical tests. Relay leads shall be formed and the mounting pad removed before the seal test. After the seal test, the mounting pad shall be rigidly attached to the relay and the remaining group A tests performed.

Qualification inspection (reduced testing) and sample size: See table III.

Figure 1 only - If the relays produced for MIL-PRF-39016/15 are similar in construction and design except for the diodes and coil assembly, as applicable, to the relays produced for MIL-PRF-39016/16, /20, or /21, then reduced testing for qualification of MIL-PRF-39016/15 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/16, /20, or /21. For reduced testing, see table III.

Figure 2 only - If the relays produced for MIL-PRF-39016/15 are similar in construction and design except for the diodes and coil assembly, as applicable, to the relays produced for MIL-PRF-39016/16, /20, or /21, then reduced testing for qualification of MIL-PRF-39016/15 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/16, /20, or /21.

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TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage - Q1 of qualification inspection table 1 unsealed sample unit for internal examination.

SUPERSESSION DATA:

Supersession data: See table IV.

TABLE IV. Supersession data. 1/

Superseded part no. M5757/99-	New part no. M39016/15- 2/	Superseded part no. M39016/15-	New part no. M39016/15- 2/	Superseded' part no. M39016/15-	New part no. M39016/15- 2/
001	034	001	029	053	065
002	029	002	030	054	066
003	030	003	031	055	067
004	031	004	032	056	068
005	032	005	033	057	069
006	033	006	034	058	070
007	040	011	041	059	071
008	035	012	042	060	072
009	036	013	043	061	073
010	037	014	044	062	074
011	038	015	045	063	075
012	039	016	046	064	076
013	046	017	035	107	101
014	041	018	036	108	102
015	042	019	037	109	103
016	043	020	038	110	104
017	044	021	039	111	105
018	045	022	040	112	106
019	052	023	047	119	113
020	047	024	048	120	114
021	048	025	049	121	115
022	049	026	050	122	116
023	050	027	051	123	117
024	051	028	052	124	118

1/ Dash numbers -029 through -034, -041 through -046, and -065 through -076 are inactive for new design and are for support of existing equipment designs only.
2/ Complete part number shall contain suffix letter L, M, P or R to designate failure rate level (see 2/ of table I). A part with any failure rate supersedes the applicable MIL-R-5757 part.

Cross reference for Government logistical support: See table V.

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TABLE V. Cross reference for Government logistical support.

Superseded part no. M5757/99-	New part no. M39016/15-	Support with part no. M39016/		Superseded part no. M39016/15-	New part no. M39016/15-	Support with part no. M39016/		New part no. M39016/15-	Support with part no. M39016/
001	034	15-034		001	029	15-029		077	15-077
002	029	15-029		002	030	20-009		078	20-051
003	030	20-009		003	031	20-010		079	20-052
004	031	20-010		004	032	20-011		080	20-053
005	032	20-011		005	033	20-012		081	20-054
006	033	20-012		006	034	15-034		082	15-082
007	040	15-082		011	041	15-029		083	15-077
008	035	15-077		012	042	20-009		084	20-051
009	036	20-051		013	043	20-010		085	20-052
010	037	20-052		014	044	20-011		086	20-053
011	038	20-053		015	045	20-012		087	20-054
012	039	20-054		016	046	15-034		088	15-082
013	046	15-034		017	035	15-077		089	15-089
014	041	15-029		018	036	20-051		090	15-090
015	042	20-009		019	037	20-052		091	15-091
016	043	20-010		020	038	20-053		092	15-092
017	044	20-011		021	039	20-054		093	15-093
018	045	20-012		022	040	15-082		094	15-094
019	052	15-082		023	047	15-077		095	15-089
020	047	15-077		024	048	20-051		096	15-090
021	048	20-051		025	049	20-052		097	15-091
022	049	20-052		026	050	20-053		098	15-092
023	050	20-053		027	051	20-054		099	15-093
024	051	20-054		028	052	15-082		100	15-094
				053	065	15-065		101	15-101
				054	066	15-066		102	20-063
				055	067	15-067		103	20-064
				056	068	15-068		104	20-065
				057	069	15-069		105	20-066
				058	070	15-070		106	15-106
				059	071	15-065		113	15-113
				060	072	15-066		114	15-114
				061	073	15-067		115	15-115
				062	074	15-068		116	15-116
				063	075	15-069		117	15-117
				064	076	15-070		118	15-118
				107	101	15-101		125	15-125
				108	102	20-063		126	20-075
				109	103	20-064		127	20-076
				110	104	20-065		128	20-077
				111	105	20-066		129	20-078
				112	106	15-106		130	15-130
				119	113	15-113			
				120	114	15-114			
				121	115	15-115			
				122	116	15-116			
				123	117	15-117			
				124	118	15-118			

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Referenced documents. In addition to MIL-PRF-39016, this document references the following:

A-A-55485, /5
MIL-PRF-39016/16, /20, /21
MIL-STD-202
MIL-STD-750
MIL-STD-1285

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NS

(Project 5945-1253)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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